THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A thick copper wire is chosen for home wiring because it: 1

(a) heats less due to low resistance **(b)** is an insulator
(c) melts easily **(d)** is a magnet

2. An electric fuse of a house is rated 15 A. Appliances of 2 A, 5 A, 6 A and 4 A are connected together. The fuse will: 1

(a) melt as 17 A exceeds 15 A **(b)** stay intact
(c) become a magnet **(d)** become hot only

3. An iron nail is replaced by a plastic rod in an electromagnet. The magnetic effect will: 1

(a) become very weak **(b)** reverse **(c)** increase **(d)** stay the same

4. A coil wound on a soft iron core is connected to a cell. The ends of the core behave like: 1

(a) two north poles **(b)** north and south poles
(c) no poles **(d)** two south poles only

5. The mains supply in India is about: 1

(a) 220 V **(b)** 9 V **(c)** 1.5 V **(d)** 6 V

6. Which one of the following would melt first in an overloaded circuit? 1

(a) copper wire **(b)** switch **(c)** bulb holder **(d)** fuse wire

7. A thin wire gets hot but a thick wire does not when the same current flows. This is because the thin wire has: 1

(a) greater resistance **(b)** less resistance
(c) no resistance **(d)** no current

8. The electrical appliances in a house are connected in: 1

(a) a spiral **(b)** series **(c)** parallel **(d)** random

9. A 5 A fuse is used for an appliance drawing 8 A. Then: 1

(a) the fuse will melt **(b)** the appliance will glow
(c) fuse will become a magnet **(d)** the appliance will be safe

10. An electric heater should be unplugged before cleaning because: 1

(a) it may cause a shock **(b)** it becomes a magnet
(c) it breaks **(d)** it makes noise

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the switch always connected to the live wire? 2

12. Why is a magnetic needle used in Oersted's experiment instead of an iron rod? 2

13. Why is a fuse always connected in series with the appliance? 2

14. Why does a bulb of a household circuit stay lit when another bulb is switched off? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how an electric bulb works. What happens inside its filament? 3

16. A bulb glows dim in a circuit with a long thin wire. Explain why. 3

17. Explain how electricity can be wasted and how it can be saved. 3

OR

A circuit has a 3 V cell and a bulb. Reverse the cell. What happens to the bulb and to a compass placed near the wire? 3

18. Differentiate between series and parallel connection of bulbs with one advantage of each. 3

OR

Describe the energy conversions in 3
(a) a heater
(b) a bell
(c) a cell.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A house circuit has a 10 A fuse. Appliances of 3 A, 2 A and 4 A are on. A 3 A appliance is added. What happens? Explain the working of the fuse and ways to avoid the problem. 5

OR

A student connects a bulb with 5
(i) a coin
(ii) a rubber eraser
(iii) salt water
(iv) a pencil lead in a gap. Predict the results and explain.

20. Compare the heating effect and the magnetic effect of electric current with two uses of each and one precaution each. 5

OR

Explain, with an example, how current, resistance and heat are related, using two wires of different thickness. 5

END OF QUESTION PAPER — Total: 40 Marks

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